UNIVERSITY OF THE PUNJAB

Third Semester 2015
Examination: B.S. 4 Years Programme

Roll No. ....

PER: Elementary Statistics
urse Code: STAT-211/

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

(SECTION 1)

Q.2: Write short answers to the following questions. (20) Marks

(i) Define Statistics.
(ii) Differentiate between sampling and non-sampling error.
(iii) Differentiate between primary and secondary data .
(iv) Define independent and dependent variable.
(v) Define Correlation and Regression.
(vi) Define Type-I error and Type-II error.
(vii) Write down two properties of correlation coefficient.
(viii) What is meant by Skewness and how it is measured.
(ix) Describe association and chi-square test .
(x) Write down any two properties of Normal Distribution.

SECTION 2

Solve the following questions: (30) Marks

Q.3: A Managing director of a large firm would like to use employee's personal characteristics to measure relationship between job performance and initiative. He took a sample of 10 employees and got the following data:

Job Performance(X)	45	65	73	63	83	45	60	73	74	69
Initiative (Y)	47	49	33	37	36	42	37	43	33	43

Compute correlation coefficient between job performance and initiative. (10) Marks

Q. 4: The following frequency distribution gives the ages of 100 college Students:

Ages	14-15	16-17	18-19	20-21	22-23	24-25	Total
Number of students	6	16	20	31	15	12	100

Calculate Arithmetic Mean, Median and Mode. (10) Marks

Q. 5: Assume that the systolic Blood Pressure of men is normally distributed with mean BP 120 mm of Hg and standard deviation 10 mm of Hg. Determine the proportion of men whose BP is above 140 mm of Hg. (5) Marks

Q.6: A restaurant management tells to its customers that the average cost of a dinner is Rs.500 with standard deviation Rs.75. A group of concerned customers think that the average cost is higher. In order to test the restaurant claims, 100 customers purchase a dinner at the store and mean price is Rs.520. Perform a hypothesis test at 1% level of significance. (5) Marks

UNIVERSITY OF THE PUNJAB

Third Semester 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Elementary Statistics — TIME ALLOWED: 2 hrs. & 30 mins
Course Code: STAT-211/GEN-21129 — MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2: Write short answers to the following questions: **(20 marks)**

(i) Define Statistics.
(ii) Differentiate between Parameter and Statistics.
(iii) Define Variance.
(iv) Write properties of the Arithmatic Mean.
(v) Define correlation.
(vi) Define Type I Error and Type II Error.
(vii) Write down two properties of Standard Deviation.
(viii) What do you mean by relative dispersion?
(ix) Describe Probability Sampling.
(x) Define additional law of probability for mutually exclusive events.

QUESTIONS WITH BRIEF ANSWERS **(30 marks)**

Q. No.3: The following frequency distribution gives the ages of 80 workers in a factory:

Ages	19-25	26-32	33-39	40-46	47-53	54-60	Total
Number of Workers	6	16	20	21	15	2	100

Compute Mean , Median and Variance. **(10 Marks)**

Q. No.4: The data of heights and weights is given below:

Height (X)	65	64	73	63	66	65	60	70	71	69
Weight (Y)	57	59	68	64	68	72	67	73	69	72

Find Regression Line of Y on X and Correlation Coefficient between X and Y. **(10 Marks)**

Q. No.5: Prepare a frequency table for the following data taking classes as 21-30, 31-40,....

70, 62, 58, 36, 56, 25, 45, 89, 45, 78, 54, 62, 42, 73, 46, 24, 39, 47, 58, 65, 43, 21, 54, 70, 56, 80, 41, 64, 26, 39, 44, 28, 33, 84, 66, 54, 22, 33, 87, 25, 55, 73, 56, 55, 70, 49, 37, 28, 46, 61, 37, 83, 47, 59, 67, 43, 29, 34, 76.

Also draw the Histogram. **(10 Marks)**

UNIVERSITY OF THE PUNJAB

Third Semester 2018
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Elementary Statistics
Course Code: STAT-211/GEN-21129

TIME ALLOWED: 2 hrs. & 3(
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2: Write short answers to the following questions: (20 marks)

i) Differentiate between Primary Data and Secondary Data.
ii) Differentiate between Parameter and Statistic.
iii) Define Relative dispersion.
iv) Define Tabulation.
v) Define Positive and Negative Correlation.
vi) Define Testing of Hypothesis.
vii) Write down two properties of Arithmatic Mean.
viii) What do you mean by Classification of data.
ix) Describe regression coefficient.
x) Define Probability Sampling.

QUESTIONS WITH BRIEF ANSWERS (30 marks)

Q. No.3: Prepare a frequency table for the following data taking classes as 11-20, 21-3, ...

100, 96, 92, 88, 86, 84, 82, 70, 62, 58, 36, 56, 25, 45, 89, 45, 78, 54, 62, 42, 73, 46, 24, 39, 47, 58, 65, 43, 15, 54, 70, 56, 80, 41, 64, 26, 39, 44, 28, 33, 84, 66, 54, 22, 33, 87, 25, 55, 73, 56, 55, 70, 49, 37, 28, 46, 61, 37, 83, 47, 59, 67, 43, 29, 34, 76.

Also draw the Histogram. (10 Marks)

Q. No.4: The following frequency distribution gives the ages of 100 college Students:

Ages	14-15	16-17	18-19	20-21	22-23	24-25	Total
Number of students	6	16	20	31	15	12	100

Compute Coefficient of Variation (C.V.). (10 Marks)

Q. No.5: The data of heights and weights is given below:

Height (X)	65	64	73	63	66	65	60	70	71	69
Weight (Y)	57	59	68	64	68	72	67	73	69	72

Find Regression Line of X on Y and Correlation Coefficient between X and Y. (10 Marks)

UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

Roll No. ....

PAPER: Elementary Statistics
Course Code: STAT-211/GEN-21129 Part – II

MAX. TIME: 2 Hrs. 45 Min.
MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Questions with short answers. **(4x5=20)**

i) Differentiate between Population and Sample.
ii) Differentiate between Parameter and Statistic.
iii) Define Dispersion.
iv) Define Classification.
v) Define Positive and Negative Correlation.
vi) Define Null Hypothesis.
vii) Write down two properties of Arithmatic Mean.
viii) What do you mean by Probability?
ix) Describe Regression.
x) Define Sampling Without Replacement.

QUESTIONS WITH BRIEF ANSWERS **(3x10=30)**

Q. No.3: A population consists of values 3, 5, 7, 9. Take all possible samples of size 2 with replacement from this population. Find the mean for each sample. Make a sampling distribution of mean and verify that:
(a) Mean of Sample Means = Population Mean
(b) Variance of Sample Means = Population Variance/n (10 Marks)

Q. No.4: The following frequency distribution gives the ages of 100 college Students:

Ages	14-15	16-17	18-19	20-21	22-23	24-25	Total
Number of students	6	16	20	31	15	12	100

Compute Mean and Standard Deviation. (10 Marks)

Q. No.5: The data of heights and weights is given below:

Height (X)	72	54	73	63	66	65	60	70	71	69
Weight (Y)	67	57	68	64	68	72	67	73	69	72

Find Regression Line of X on Y and Correlation Coefficient between X and Y. (10 Marks)

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Third Semester – Fall 2021

Roll No. ....

Paper: Elementary Statistics   Course Code: STAT-211   Time: 3 Hrs. Marks: 60

Q.1. Answer the following short questions: **(15x2=30)**

i) Define Statistics.

ii) Define population and sample.

iii) Define attribute and give an example.

iv) Define Classification.

v) What is Measure of Central Tendency?

vi) Define the following (a) Median (b) Mode

vii) What is Dispersion?

viii) Define Variance.

ix) Write advantages of sampling.

x) Define Probability Sampling.

xi) Define Type I Error and Type II Error.

xii) Define Critical region.

xiii) Define Simple Regression.

xiv) Define independent and dependent variable in a Simple Regression.

xv) Write down two properties of Correlation Coefficient.

Answer the following questions. **(3x10=30)**

Q2. The weights of the 40 male students at a university are given in the following frequency table:

Weight	118-126	127-135	136-144	145-153	154-162	163-171	172-180
Frequency	3	5	9	12	5	4	2

Compute Mean, Median, Mode and Coefficient of Variation.

Q3. A population consists of values 0,3,6,9. Take all possible samples of size 2 without replacement from this population. Find the mean for each sample. Make a sampling distribution of sample mean and verify that:

(a) $\mu_{\bar{x}} = \mu$ (b) $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\frac{N-n}{N-1}$

Q4. Calculate the coefficient of correlation and obtain the regression line Y on X for the following data:

Price (X)	3	4	5	6	7	8	9	10	11	12
Demand (Y)	67	57	68	64	68	72	67	73	69	72

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Spring 2022

Roll No. ....

Paper: Elementary Statistics **Course Code: STAT-211** **Time: 3 Hrs. Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: **(15x2=30)**

i) Define Inferential Statistics.

ii) Define Primary Data and Secondary Data.

iii) Define Classification.

iv) Define Tabulation.

v) Define the following (a) Mean (b) Mode

vi) Define Quantiles.

vii) Define Variance.

viii) Write down the two properties of Variance.

ix) Define Sampling with replacement.

x) Define the following (i) Sample (ii) Sampling frame

xi) What is level of significance?

xii) Define one tailed and two tailed test.

xiii) Define independent and dependent variable in a Simple Regression.

xiv) Define Regression Coefficient.

xv) Write down two properties of correlation coefficient.

Answer the following questions. **(3x10=30)**

Q2. The following frequency distribution gives the ages of 100 college Students:

Ages	14-15	16-17	18-19	20-21	22-23	24-25	Total
Number of Students	6	16	20	31	15	12	100

Compute Mean, Standard Deviation and Coefficient of variation.

Q3. The data of heights and weights is given below:

Height (X)	72	54	73	63	66	65	60	70	71	69
Weight (Y)	67	57	68	64	68	72	67	73	69	72

Find Regression Line of Y on X and Correlation Coefficient between X and Y.

Q4. Draw all possible samples of size n=2 with replacement from the population 3, 6, 9 and 12. Find the proportion of even number in the samples. Form a sampling distribution of the sample proportion. Verify that:

(a) $\mu_p = P$ (b) $\sigma_p^2 = \frac{P(1-P)}{n}$

where p is sample proportion and P is population proportion.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Third Semester – Spring 2023

Roll No. ....

Paper: Elementary Statistics | **Course Code: STAT-211** | **Time: 3 Hrs. Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (15x2=30)

I. Differentiate between descriptive statistics and inferential statistics.
II. Name at least 4 sources of primary data.
III. What is the difference between a histogram and a bar chart?
IV. Differentiate between population and sample.
V. Differentiate between symmetry and skewness.
VI. Differentiate between median and mode.
VII. Define the term "variance" and explain how it is calculated.
VIII. Differentiate between sampling error and non-sampling error.
IX. Differentiate between one-way classification and two-way classification.
X. Define the term probability.
XI. What is the difference between probability sampling and non-probability sampling?
XII. Define the term Poisson distribution.
XIII. Define simple correlation.
XIV. Define normal distribution and write down its two properties.
XV. Explain type-1 error with one example.

Answer the following questions. (3x10=30)

Q2. Draw all possible samples of size 2 with replacement from a population consisting of 3, 6, 9. Compute sampling distribution of sample means and verify:

(i) $E(\bar{X})=\mu$ (ii) $Var(\bar{X})=\frac{\sigma^2}{n}$

Q3. The height and weight of 10 persons are given in the following table. Estimate the correlation coefficient between X and Y.

Height (X)	45	46	47	48	49	50	51	52	53	54
Weight (Y)	110	111	111	115	116	118	120	121	124	128

Q4. Compute the Coefficient of Variation of the following data:

Classes	14-16	17-19	20-22	23-25	26-28	29-31	32-34
Frequency	7	9	3	8	6	4	2

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Spring 2024

Roll No. 215334

Paper: Elementary Statistics | Course Code: STAT-211 | Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: **(15x2=30)**

i) Define Descriptive Statistics.

ii) Differentiate between Parameter and Statistic.

iii) Define Tabulation.

iv) Define Frequency Distribution.

v) Write down two properties of Arithmetic mean.

vi) Write empirical relationship between mean, median and mode for skewed distribution.

vii) Define Standard Deviation.

viii) What is meant by Skewness and write one method to measure it.

ix) Define Non Probability Sampling.

x) Differentiate between Sampling error and Non-Sampling error.

xi) What is Testing of Hypothesis?

xii) Define Test Statistic.

xiii) Define Regression.

xiv) Define Coefficient of Correlation.

xv) Write down two properties of regression line.

Answer the following questions: **(3x10=30)**

Q2. The following are the scores made by two batsmen A and B in a series of innings:

A	12	15	6	73	7	19	199	36	84	29
B	47	12	76	48	4	51	37	48	13	0

Who is better as a run getter? Who is more consistent player?

Q3. A population consists of values 3,5,7,9. Take all possible samples of size 2 with replacement from this population. Find the mean for each sample. Make a sampling distribution of sample mean and verify that:

(a) $\mu_{\bar{x}} = \mu$ (b) $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$

Q4. Find regression line X on Y and a Coefficient of Correlation between X and Y, given:

X	5	12	4	16	18	21	22	23	25
Y	47	12	76	48	4	51	37	48	13